

CF3 Sample Restrictions and Nuisance-Adjustment Appendices

English Manuscript Version

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Appendix B. English Manuscript Version

Appendix Table A1. Rationale, implementation, and audit of sample restrictions and nuisance adjustments

Category	Item	Implementation	Rationale	Principal limitation	Audit result
Baseline	Unadjusted association	Pearson, Spearman, and log–log regression of cataloged K_s -band luminosity against H_{loc}	Establish whether the positive sign is already present before adjustment	Does not address redshift selection, distance quality, catalog correction, or sky-dependent effects	Pearson $r = 0.0844$, $p = 0.0134$; Spearman $\rho = 0.0840$, $p = 0.0138$. The positive sign is therefore not created by adjustment
Eligibility	Complete required fields	Required finite distance, CMB-frame velocity, luminosity, richness, correction factor, distance-count, and distance-error fields	Prevent missing or invalid quantities from entering logarithms and regressions	Complete-case selection can reduce representativeness	6,606 of 11,508 table-2 groups had all required fields
Sample restriction	Velocity interval	$3000 \leq v_{\text{CMB}} \leq 15000 \text{ km s}^{-1}$	Reduce dominance by nearby peculiar velocities and limit strong high-distance catalog selection	Restricts external validity; thresholds were not prospectively preregistered	Alternative plausible velocity intervals retained a positive adjusted coefficient
Sample restriction	Distance-measurement count	$N_{\text{dist}} \geq 2$	Exclude systems relying on a single group-distance measurement	Measurement count may correlate with luminosity, distance, and observational attention	Adjusted coefficients remained positive for thresholds of one, two, or three distance measurements
Sample restriction	Distance-modulus uncertainty	$\sigma_{\mu} \leq 0.30 \text{ mag}$	Limit distance noise because D enters directly into $H_{\text{loc}} = v_{\text{CMB}}/D$	The threshold is partly conventional and may introduce quality-related selection	Positive signs were retained under 0.20, 0.30, and 0.40 mag thresholds
Sample restriction	Luminosity correction factor	$1 \leq C_F \leq 10$	Exclude systems requiring extreme missed-light corrections	C_F is strongly related to cataloged luminosity and affects model sensitivity	Only one additional group was removed at this stage; alternative plausible C_F limits retained a positive sign
Sample restriction	H_{loc} interval	$40 \leq H_{\text{loc}} \leq 110 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Exclude extreme velocity–distance ratios	This is an outcome-based restriction and can, in principle, induce selection bias	Only two groups were excluded. Without this restriction, $N = 860$, $\beta = 0.03808$, 95% CI 0.00606–0.07009, and $p = 0.01997$, essentially unchanged from the primary result

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Nuisance adjustment	Velocity trend	Centered linear, quadratic, and cubic terms in $\log_{10} v_{\text{CMB}}$	Model nonlinear redshift-, distance-, and flux-selection trends	Because v_{CMB} also enters H_{loc} , the adjusted outcome is not a pure metric-expansion measure	With the cubic velocity function alone, $\beta = 0.01559$ and $p = 0.000279$
Nuisance adjustment	Selection-correction factor	Included $\log_{10} C_F$	Control catalog-level missed-light and selection corrections	Statistical significance is sensitive to this control	Omitting C_F from the full model retained a positive coefficient, $\beta = 0.02813$, but gave $p = 0.168$
Nuisance adjustment	Group richness	Included $\log_{10} N_{\text{pv}}$	Compare luminosity differences among systems of similar cataloged richness	Richness may partly mediate the total-luminosity signal and therefore changes the estimand	Without richness adjustment, the result remained positive: $\beta = 0.01624$, $p = 0.0288$
Nuisance adjustment	Distance quality	Included $\log_{10} N_{\text{dist}}$ and σ_{μ}	Control heterogeneity in the number and precision of distance measurements	Does not remove distance-indicator calibration differences or measurement error completely	Omitting the distance-quality block gave $\beta = 0.04019$, $p = 0.0118$
Nuisance adjustment	Sky direction	Included the dipole components ($\cos b \cos l, \cos b \sin l, \sin b$)	Control first-order sky coverage, calibration, and directional-flow structure	A dipole adjustment cannot reconstruct the full peculiar-velocity field	Omitting the sky block gave $\beta = 0.04149$, $p = 0.0114$
Full model	All nuisance blocks	Simultaneous adjustment for the cubic velocity function, C_F , richness, distance quality, and sky dipole	Estimate the luminosity association conditional on the principal measured catalog trends	Nuisance variables explain approximately 95% of luminosity variance; the luminosity VIF is approximately 20, making the magnitude model-sensitive	$\beta = 0.03837$, HC3 95% CI 0.00677–0.06997, $p = 0.01755$
Uncertainty	HC3 standard errors	Heteroskedasticity-consistent HC3 covariance estimator	Avoid underestimated uncertainty under nonconstant residual variance	Corrects the estimated covariance but not confounding or model misspecification	Full-model HC3 SE = 0.01612
Robustness	Nonparametric bootstrap	3,000 row-resampling replicates with a fixed random seed	Assess coefficient uncertainty without relying solely on normal residual assumptions	Remains conditional on the observed sample and selected model	Bootstrap 95% CI 0.00794–0.06883
Robustness	Residual Spearman test	Rank correlation between luminosity and H_{loc} residuals after the common nuisance projection	Reduce dependence on the OLS scale and sensitivity to extreme observations	Remains a conditional association, not a causal estimate	$\rho = 0.0850$, $p = 0.0128$
Robustness	Stratified permutation	Permuted luminosity residuals within 20 equal-count velocity strata for 5,000 replicates	Test whether the residual association could arise under random rearrangement while preserving velocity structure	Does not address unmeasured confounding and depends on the selected stratification	$p_{\text{perm}} = 0.0144$
Sensitivity sample	Strict-quality subset	Required $N_{\text{dist}} \geq 3$, $\sigma_{\mu} \leq 0.20$ mag, and $C_F \leq 5$	Examine the association under stronger distance-quality and correction requirements	Nested within the primary sample and therefore not an independent replication	$N = 242$, $\beta = 0.0963$, $p = 0.00468$; the larger estimate demonstrates magnitude sensitivity
Sensitivity sample	SN Ia-only subset	Constructed distances from the median repeated SN Ia distance modulus within each group	Test the association using a more homogeneous distance-indicator subset	Smaller sample and not fully independent of the catalog construction	$N = 188$, $\beta = 0.0106$, $p = 0.435$; the primary association was not reproduced

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Visualization	Six equal-count bins	Divided the 858 nuisance-adjusted luminosity residuals into six equal-count bins; all regressions still used the 858 individual groups	Provide a readable visual summary of the weak conditional association	Binning suppresses individual scatter and can make the trend appear stronger than the underlying correlation	The bins are descriptive only and do not determine the coefficient, confidence interval, or p value

Table note suitable for insertion:

$H_{\text{loc}} = v_{\text{CMB}}/D$ is an observer-frame redshift–distance proxy and not a direct covariant expansion scalar. Adjustment removes modeled catalog trends only; it does not reconstruct peculiar velocities, eliminate unmeasured confounding, or establish causation. The analysis was exploratory and was not prospectively preregistered. Consequently, the stability of the positive sign is more defensible than any precise interpretation of the coefficient magnitude or response function.